

ANUJ GAJJAR

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WORK EXPERIENCE

THE MITRE CORPORATION

Sep 2022 – Present

Senior Embedded Software Engineer – *C++, C, Python, Node.js, Vue.js, Linux, FreeRTOS, Docker, CMake, Git* Bedford, MA

- Developed a multi-threaded, event-driven C++17 GNSS/GPS system on Linux supporting multiple applications
- Owned development of core components spanning hardware/FPGA drivers, embedded platform bring-up, signal processing pipelines, telemetry, data encoding/decoding, cryptography-related functionality, and CI infrastructure
- Developed networking and serial interfaces such as Ethernet, TCP/IP, I2C, SPI, and UART
- Developed Python tooling, CLI interfaces, and a Vue.js web dashboard for debugging, real-time monitoring, and analysis
- Optimized CPU and memory usage to meet real-time performance constraints
- Resolved complex concurrency and timing issues, enabling stable multi-month system deployments
- Resolved hardware and signal-level issues using spectrum analyzers and oscilloscopes
- Containerized build, test, and release workflows using Docker, reducing deployment time by 3+ hours
- Authored user and developer documentation and provided technical support to cross-functional users and teams

UNIVERSITY OF NOTRE DAME

May 2021 – July 2021

Advanced Wireless Research Experience (AWARE) Fellow – *C++, Python, CMake, Git*

Notre Dame, IN

- Built Python and C++ libraries for simulating, testing, and characterizing an implantable smart breast clip that provides real-time information on the condition of breast cancer tumors (presented at SPIE BiOS, Campeau et al., 2024)

NORTHEASTERN UNIVERSITY – Optical Science Laboratory

Jul 2019 – May 2022

Undergraduate Researcher – *Python, MATLAB, PyTorch, OpenCV*

Boston, MA

- Developed convolutional neural nets and optical simulations with PyTorch and MATLAB to determine the orientation of collagen molecules (published in Journal of Biomedical Optics, Alzola et al., 2021)
- Prototyped a machine learning-based image translation technique using OpenCV and PyTorch to convert confocal images of human skin, reducing microscopy acquisition time for skin cancer and disorder diagnosis

EDUCATION

NORTHEASTERN UNIVERSITY

Sep 2018 – May 2022

B.S. in Electrical and Computer Engineering – GPA 3.8/4.0

Boston, MA

Activities: Sherman Center for Entrepreneurship, Generate Product Development

Coursework: Embedded Design, Operating Systems, Wireless Communication Circuits, Medical Imaging, Electromagnetics

Publications: Assessing Tissue Interrogation Volume of an Implantable Optical Sensor (SPIE BiOS, Campeau et al., 2024)
Measurement of Collagen Monomer Orientation (Journal of Biomedical Optics, Alzola et al., 2021)

PROJECTS

DISCTRACKER – *C++, FreeRTOS, Swift/iOS, Electronics & PCB design*

- Developed a functional IoT prototype for disc golf using custom electronics/PCB design and a FreeRTOS-based embedded system that streamed gameplay metrics
- Built an iOS companion application in Swift to visualize real-time data gathered over Bluetooth

MODE Radar – *C, TI RTOS, Python, Electronics & PCB design*

- Architected a millimeter-wave radar system achieving <10mm resolution for disaster environment mapping
- Designed a custom PCB and RF phased array antenna, and implemented low-level control software in C using TI RTOS
- Developed multi-process Python software for real-time data processing and communication

ShowNXT – *React Native, Node.js, GraphQL, PostgreSQL, Git*

- Developed a full-stack recruitment platform for college athletics using React Native, GraphQL, and PostgreSQL
- Implemented core features including real-time in-app chat, user profile management, and a media-rich interface for uploading and viewing athlete highlight reels